

Fig. 6: Actual-size PCB of receiver

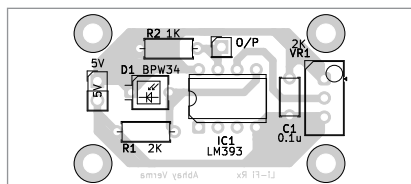


Fig. 7: Components layout of receiver PCB

location for simplicity.

At the receiver end, power is obtained directly from the power rails of the USB-to-TTL converter module, eliminating the need for an external battery supply. A Python script (Fig. 5) is implemented on the receiver system to receive the incoming optical data and reconstruct it into the original file. The program continuously monitors the specified serial port at 260000 baud.

After the serial connection is initialised and stabilised, the receiver waits for a marker to synchronise with the transmitter. Once synchronisation is established, the program reads the incoming metadata, including the total file size and packet size, which are used to control the reception and reconstruction process. A new output file is then created in binary write mode to store the received data.

The receiver program enters a continuous loop in which it reads incoming packets beginning with a fixed header (0xAA 0x55) to maintain proper synchronisation and packet alignment. Each packet is decoded into its individual fields, including the packet number, data length, payload, and checksum. The checksum is recalculated using the same method as the transmitter (sum of bytes modulo 256) and compared with the received checksum to verify data integrity. If the checksum matches, the payload data is written to the output file; otherwise, a CRC error message is displayed, indicating corrupted data.

The reception process continues until the received data matches the expected file size, ensuring accurate reconstruction of the transmitted file. After the transfer completes successfully, the reconstructed file is saved automatically, and the serial connection is closed safely. Fig. 5 shows a snippet of the Python receiver program used for Li-Fi optical data reception and file reconstruction.

Construction and testing

The transmitter and receiver circuits can be assembled easily on a breadboard. A dedicated PCB for the transmitter was not designed due to the circuit's simplicity. The actual-size PCB of the receiver is shown in Fig. 6, and its component layout is shown in Fig. 7.

After assembling the circuit on the PCB, mount it inside a suitable enclosure with an opening for the laser beam so that the light falls directly on the photodiode. Connect the PCB to a laptop or desktop computer through the USB interface.

For testing, the data transfer speed was set to about 260kbps, the maximum stable speed achieved after optimising all components. At this speed, the bit-switching time is nearly 4µs, which is close to the practical operating limit of the LM393 comparator. Increasing the baud rate further caused slower comparator response, resulting in corrupted data and CRC errors.

Capacitor C1 in the receiver circuit plays an important role in signal conditioning and filtering, enabling stable operation at 260000 baud. Without C1, the maximum

PARTS LIST

Semiconductors:

- LD1 - Laser diode
- T1 - BC547 NPN transistor
- D1 - BPW34 PIN photodiode
- IC1 (IC1A-IC1C) - LM393 comparator IC

Resistors (all 1/4-watt, ±5% carbon):

- R1-R3 - 1kΩ
- VR1 - 2kΩ preset

Capacitor:

- C1 - 0.1µF ceramic disc capacitor

Miscellaneous:

- U1, U2 - CH9102 USB-to-TTL converter modules
- General-purpose PCB/breadboard
- Connecting/jumper wires
- Breadboard
- Wires and jumpers (as required)
- Enclosure (optional)

reliable speed was limited to around 200000 baud.

Resistor R1 controls the balance between the photodiode stage's sensitivity and noise immunity. Higher R1 values improve immunity to ambient light noise but reduce sensitivity, while lower values improve sensitivity and response speed at the cost of greater noise susceptibility.

Learning points

The system demonstrates how digital data can be transmitted using light and received through a photodiode using the on-off keying (OOK) technique. It also shows how data packets containing headers, payload, and checksums enable reliable file transfer. The design highlights the importance of comparators, filtering circuits, and proper component selection in improving signal quality and noise immunity. In addition, the system provides a practical understanding of high-speed UART communication and hardware-software interfacing. Overall, the system illustrates how discrete electronic circuits and software algorithms can be combined to build a complete Li-Fi optical communication system. **EFY**

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